SIMBa

BGSMath
BARCELONA GRADUATE SCHOOL OF MATHEMATICS

Seminari Informal de Matemàtiques de Barcelona

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University: Universitat Politècnica de Catalunya.

Date: Wednesday, June 3rd, 2026.
Schedule: 12:20 talk.
Place: UPC (Facultat de Matemàtiques i Estadística-S01) and Zoom.
Language: English.

Title: Predator-prey systems: comparison of functional responses in ecology and applications to bacteriophage-bacteria systems.

Abstract: Describing the interaction between predators and prey is one of the classical problems in mathematical ecology. A central component of these models is the functional response, which describes how the predator consumption rate depends on prey density.
Different ecological mechanisms –such as prey clustering, handling efficiency, or predator interference– can lead to substantially different functional responses and, consequently, to different dynamical behaviours. In this talk, we present and compare several classical predator-prey models, beginning with Holling functional responses of types I, II, and III, and extending to Beddington-DeAngelis and non-monotone Holling type IV models associated with herd behaviour. We discuss the biological assumptions underlying each formulation and analyse how these modify the qualitative dynamics of the system. We then apply these ideas to a SIR-type bacteriophage-bacteria interaction ODE model, where bacteriophages act as predators and bacteria as prey. We consider both planktonic and biofilm environments, showing how spatial structure and environmental effects naturally lead to different functional responses. Finally, we discuss equilibria and near-equilibrium behaviour in these models. We focus on parameter regimes where phage production and phage loss nearly balance, leading to long transient dynamics that may appear as biologically relevant pseudo-equilibria over experimental time scales.

About us: *SIMBa* is a mathematics seminar organized by graduate students in the Barcelona area. It is aimed towards graduate and last–year undergraduate students. Our goals are disseminating knowledge from different branches of mathematics for those interested and promoting networking between the attendants.
This seminar is backed by the Faculty of Mathematics and Computer Science at Universitat de Barcelona, Faculty of Mathematics and Statistics at Universitat Politècnica de Catalunya, the Department of Mathematics at Universitat Autònoma de Barcelona, CRM, IMUB and BGSMath.

Fore more information, visit seminari-simba.github.io/en. You may contact us by sending an email to ***seminari.simba@gmail.com***.